

Secure Exit



SIMATS Online Programming Lab

JAVA PROGRAMMING



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QUESTIONS

Q1

Banking Transaction System –
Exception Handling

10 marks

Q2

Hospital Patient Registration –
User-Defined Exception

10 marks

Q3

E-Commerce Order Processing
– Built-in Exceptions

10 marks

Q4

Hospital Emergency System –
Thread Priority

10 marks

Q5

Online Food Delivery – Multiple
Threads and Execution Order

10 marks

Q4 Hospital Emergency System – Thread Priority

10 marks

A hospital application creates threads for Emergency Patient Processing, General Patient Processing, and Billing Processing.

Develop a Java program that assigns different priorities to these threads. Run the program several times and analyze whether the emergency thread is guaranteed to execute first.

Then modify the design if the hospital requires a strict execution sequence rather than merely expressing scheduling preference.

Your answer must contain Java code and an execution analysis based on the actual thread behavior.

Sample Test Cases:

Input / Scenario Expected Result Analysis Focus

Emergency, General, Billing All threads execute; priority does not guarantee a fixed output order

Normal execution

Emergency workload > General workload Emergency thread receives a higher priority but exact ordering is not guaranteed Concurrency / design

Strict emergency-before-billing requirement Use explicit coordination rather than relying only on priority Exception / boundary

5/5 answered

Your Code

```

1 class Main
2 {
3     static String b=null;
4     static synchronized void produce(String s) throw
5     {
6         while(b!=null)
7         {
8             Main.class.wait();
9         }
10        b=s;
11        System.out.println("Produced "+s);
12        Main.class.notify();
13    }
14    static synchronized void consume() throws Except
15    {
16        while(b==null)
17        {
18            Main.class.wait();
19        }
20        System.out.println("Consumed "+b);
21        b=null;
22        Main.class.notify();
23    }
24    public static void main(String[] args) throws Exc
25    {
26        Thread t1=new Thread(() ->
27            {
28                try
29                {
30                    produce("P1");
31                }

```

Input

stdin input

Output

Produced P1
Consumed P1
Produced P2
Consumed P2
Produced P3
Consumed P3

```
32         catch (Exception e)
33         {}
34     });
35     Thread t2 = new Thread(() ->
36     {
37         try
38         {
39             consume();
40         }
41         catch (Exception e)
42         {}
43     });
44     t1.start();
45     t2.start();
46     t1.join();
47     t2.join();
48 }
49
50 }
```



✓ Submitted